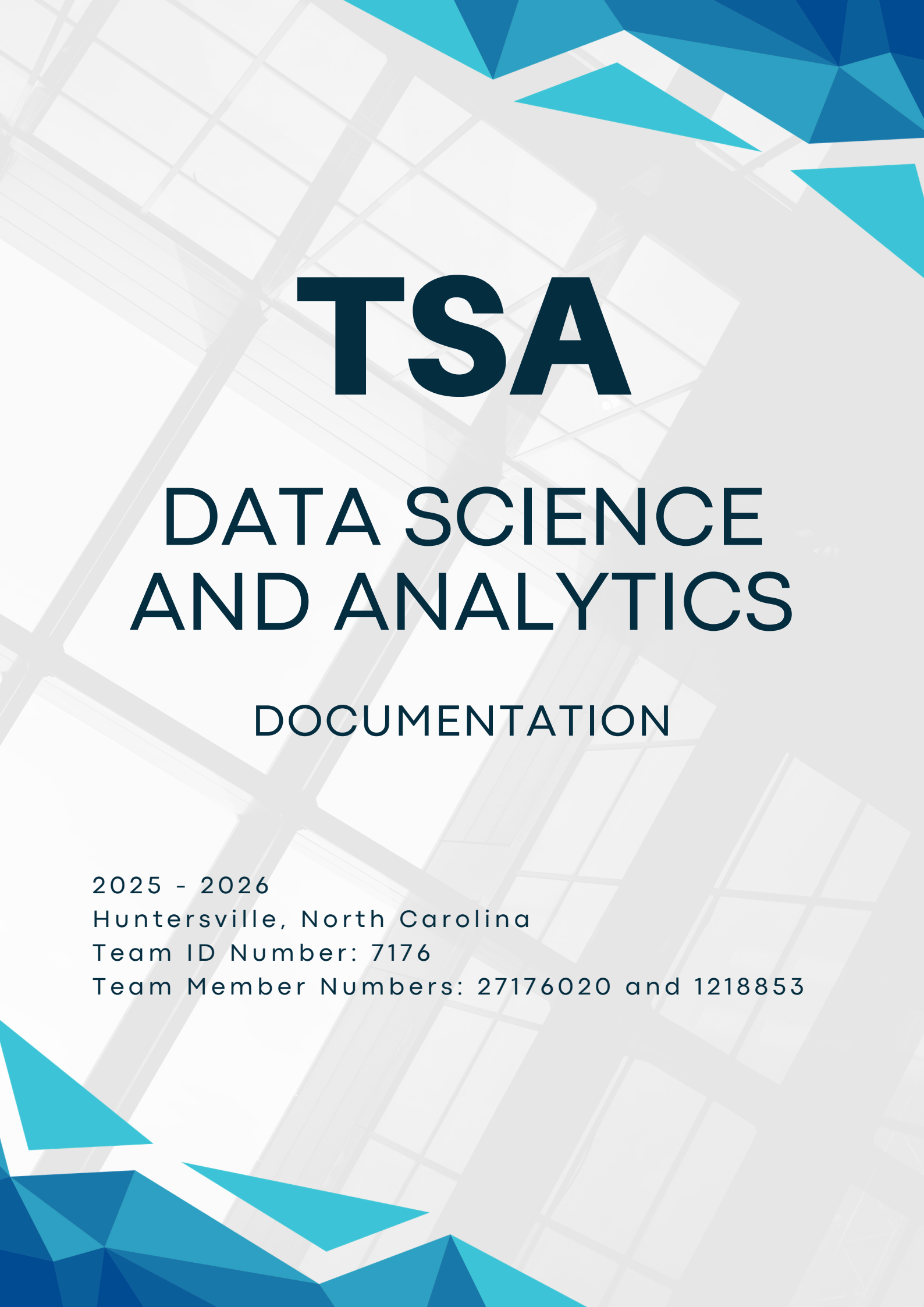




TSA

DATA SCIENCE AND ANALYTICS

DOCUMENTATION

2025 - 2026

Huntersville, North Carolina

Team ID Number: 7176

Team Member Numbers: 27176020 and 1218853

STUDENT COPYRIGHT CHECKLIST (for students to complete and advisors to verify)

STUDENT: Answer question 1 below.

- 1) Does your solution to the competitive event integrate any type of music and/or sound? ☐ YES ☒ NO

If NO, go to question 2.

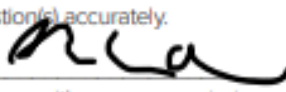
If YES, is the music and/or sound copyrighted? ☐ YES ☐ NO

If YES, move to question 1A. If NO, move to question 1B.

1A) Have you asked for author permission to use the music and/or sound in your solution and included that permission (letter/form) in your documentation? If YES, move to question 2. If NO, ask for permission and if permission is granted, include the permission in your documentation.

1B) Is the music/sound royalty free, or did you create the music/sound yourself? If YES, cite the royalty free music/sound OR your original music/sound properly in your documentation.

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to the use of music/sound in his/her competitive event solution. Even if your student answers "NO" to question 1, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I,  (chapter advisor), have checked my student's solution and confirm that any use of music/sound is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no music/sound included.

STUDENT: Answer question 2 below.

- 2) Does your solution to the competitive event integrate any graphics/videos? ☒ YES ☐ NO

If NO, go to question 3.

If YES, is(are) the graphics/videos copyrighted, registered and/or trademarked? ☐ YES ☒ NO

If YES, move to question 2A. If NO, move to question 2B.

2A) Have you asked for author permission to use the graphics and/or videos in your solution and included a permission (letter/form) in your documentation for graphic/video used? If YES, move to question 3. If NO, ask for permission and if permission is granted, include the permission in your documentation.

2B) Is(are) the graphics/videos royalty free, or did you create your own graphic? If YES, cite the royalty free graphics/videos OR your own original graphics/videos properly in your documentation.

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I,  (chapter advisor), have checked my student's solution and confirm that the use of graphics/videos with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have no graphics/videos included.

STUDENT: Answer question 3 below.

- 3) Does your solution to the competitive event use another's thoughts or research? ☒ YES ☐ NO

If NO, this is the end of the checklist.

If YES, have you properly cited other's thoughts or research in your documentation? ☒ YES ☐ NO

CHAPTER ADVISOR: Sign below regarding your student's answer(s) to having integrated any thoughts/research of others in his/her competitive event solution. Even if your student answers "NO" to question 3, please sign below noting that you have evaluated the competitive event solution and the student answered the question(s) accurately.

I,  (chapter advisor), have checked my student's solution and confirm that the use of the thoughts/research of others is done so with proper permission and is cited correctly in the student's documentation and/or the solution has been found to have all original thought with no use of other's thoughts/research.

Sanvi Nethikunta and Hitee Kakarla

Student Initials: _____

Chapter Advisor Signature: _____



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Introduction

Tourism is the traveling of people to destinations outside of where they live for entertainment or just the will to explore. Tourism plays an important role in many cities because it supports local businesses, creates jobs, and generates revenue for local governments. However, research has shown that high levels of tourism can also place pressure on urban environments and negatively affect residents' quality of life, especially in large cities (Koens et al., 2018; Mihalic, 2020).

In populated urban areas such as Boston, Seattle, and Washington DC, increased tourism activity contributes to resident service burden, which is the added strain placed on city services and infrastructure by visitor demand. This burden often appears through higher levels of noise, overcrowding in public spaces, sanitation concerns, and transportation congestion. These effects are commonly described in tourism research as tourism spillovers, where visitor activity creates unintended impacts on local residents and city services.

One commonly used concept in tourism studies is tourism intensity, which describes the level of tourism activity relative to the size of the local population. When tourism intensity becomes very high, cities may experience tourism saturation, also referred to as overtourism. Overtourism occurs when visitor demand exceeds a city's ability to manage tourism without negatively affecting residents' daily lives (Mihalic, 2020).

Tourism impacts are often assumed to increase in a simple, linear way as tourism activity grows. It varies widely across cities due to differences in infrastructure capacity, tourism management, and local conditions. As a result, similar increases in tourism activity may lead to very different levels of resident burden in different cities.

This project examines whether tourism activity is associated with increases in resident service burden and whether this relationship differs across cities. By combining Google Trends data as a measure of tourism activity with 311 service request data as an indicator of resident burden, the analysis explores how tourism pressure interacts with urban quality of life over time.

Purpose

The goal of this paper is to investigate the relationship between tourism activity in big cities and the increased resident-reported service burden, as well as whether the relationship varies across cities in order to test the assumption that increased tourism is associated with increased service/resident burden.

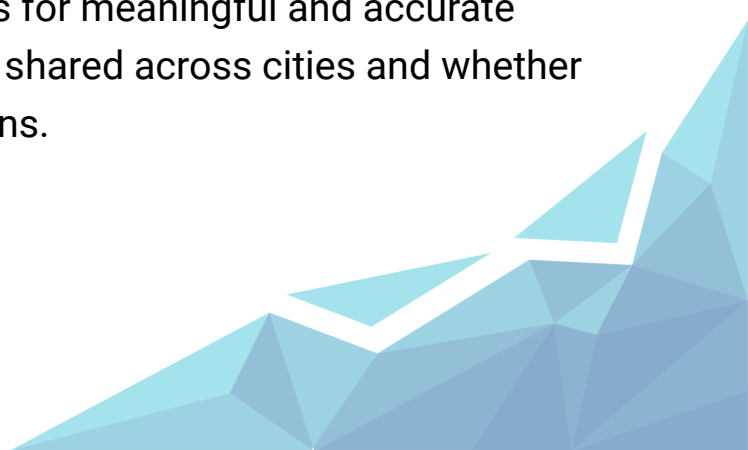
This paper uses data collection, data preparation and feature construction, quantitative analysis, and visualization and tools in order to explore this relationship.

The main objective is to analyze Google Trends and 311 service request data from city open-data portals and compare tourism activity values to tourism complaint share values for each city in order to observe how this relationship changed over time and whether trends appeared across different cities.

Three major U.S. cities this study focuses on are Boston, Seattle, and Washington, DC. These cities experience a significant amount of tourist activity and they have a 311 service request system that is publicly accessible, making them excellent examples to analyze. Boston is useful for examining economic changes and identifying trends because it consists of dense urban areas and experiences seasonal tourism (City of Boston, 2024).

Seattle is growing very rapidly and experiences lots of tourism as well as environmental issues such as crowding and noise pollution (City of Seattle, 2024). Washington, DC attracts visitors year-round due to national landmarks and events, which provides a useful comparison to more seasonal tourism patterns. (District of Columbia Government).

Analyzing data from different cities allows for meaningful and accurate results which determine broader patterns shared across cities and whether the findings are consistent across locations.



Data Overview

This project uses two primary datasets to analyze the relationship between tourism activity and resident service burden across three major cities. The data sources used were publicly accessible and provided different perspectives on tourism demand and impacts of tourism reported by residents.

Tourism Activity Data

Google Trends was used to measure tourist activity by reporting the relative popularity of specific search terms over time. Higher popularity values showed a greater search interest, and terms related to tourism were selected, representing public interest in tourist activities within each city. Monthly data was collected in order to match resident service data.

Resident Service Burden Data

311 service request records from open-data portals were used in order to measure resident service burden. These records document non-emergency issues reported by residents, such as noise disturbances, sanitation concerns, and public space disruptions, or other negative impacts on the quality of life. This study only used the complaint type and submission date. Service requests related to tourism activity were identified using an already existing set of tourism-related keywords. The data was then set to a monthly city-level format, allowing consistent comparison with tourism activity data.

Cities and Time Frame

This analysis includes three major U.S. cities to allow for comparison across locations and all data was standardized to a city-month level. This study covered multiple years, allowing for the observation of seasonal tourism patterns and long-term trends.

Final Dataset Structure

After cleaning, grouping the data into monthly values, and merging the datasets, the final dataset consists of monthly observations for each city and includes:

- a measure of tourism activity (Google Trends search interest),
- total resident service requests,
- tourism-related service requests, and
- the share of resident complaints associated with tourism activity.

This standardized structure enables consistent comparison across cities and supports the analysis of how tourism pressure and resident service burden evolve over time.

DATA DICTIONARY

The following table describes the fields used in the final analysis dataset. Each observation represents a city-month and combines tourism activity data with resident service complaint data.

Field Name	Data Type	Description	Example
city	String	Name of the city associated with the observation. Used to group and compare results across different urban environments.	"Seattle"
month	Date (YYYY-MM-DD)	Monthly time period for the observation, derived from service request timestamps and aligned with Google Trends data.	2025-07-01
n_311_calls	Integer	Total number of 311 service requests submitted by residents in the city during the given month. Represents overall resident service burden.	18,423
n_tourism_calls	Integer	Number of 311 service requests in the city-month classified as tourism-related based on complaint category keyword matching.	932

tourism_call_share	Float	Proportion of total 311 service requests that are tourism-related, calculated as tourism_311_calls divided by total_311_calls. Used to normalize complaint levels across cities.	0.0506
tourism_intensity	Integer (0–100)	Monthly Google Trends search interest value representing tourism activity in the city. Values are normalized by Google Trends and indicate relative tourism demand over time within the city.	72

Notes on Variable Construction

- Tourism-related complaints were identified using keyword matching on complaint types associated with tourism spillovers (such as noise, crowding, sanitation, transportation).
- Tourism complaint share (stored as tourism_call_share in the dataset) was used instead of raw complaint counts to allow fair comparison between cities with different population sizes and baseline complaint volumes.
- Tourism intensity values were interpreted comparatively over time within the same city rather than as absolute levels across cities.

Methods

To examine the relationship between tourism activity and resident service burden, this project followed a structured data analysis process. The methodology used consists of four main stages: data collection, data preparation, quantitative analysis, and visualization.

1. Data Sources

This project combines two publicly available datasets to examine the relationship between tourism pressure and resident service burden across major U.S. cities.

Using Google Trends, a tool that analyzes the popularity of top search queries, tourism activity was measured. To do this, tourism-related keywords were selected in order to represent interest in visiting, staying in, or engaging in activities within each city. Google Trends reports monthly search interest on a normalized scale from 0-100, where 100 represents peak relative interest for that city during the selected time range. These values were used as a proxy for tourism demand over time.

Using city 311 service request data, resident service burden was measured. This data was obtained from city open-data portals and datasets record non-emergency complaints submitted by residents about every day concerns relating to quality-of-life, such as noise, sanitation, congestion, and limited public space. For consistency across cities, only the complaint type and date fields were retained.

2. Data Preparation

All datasets were standardized to a monthly, city-level time scale to allow for direct comparison between tourism activities and resident complaints.

311 service request records were processed and converted into a standard date format. Each complaint was labeled as either tourism-related or non-tourism-related using a predefined keyword-based filtering approach. Keywords were selected to capture common tourism spillover effects, such as noise, nightlife activity, crowding, transportation congestion, and short-term visitor accommodations. The complete list of keywords used for classification is included in Appendix B.

For each city and month, complaint data was aggregated in order to produce monthly totals. Google Trends data was downloaded at a monthly frequency and aligned to the same time range as the 311 data. Months with missing or incomplete data in either dataset were removed to ensure valid comparisons.

3. Feature and Variable Construction

Three primary variables were constructed for analysis:

- **Tourism Intensity:** Monthly Google Trends search interest value for tourism-related queries within a city.
- **Total 311 calls (service requests)** Monthly count of all 311 service requests filed within a city.
- **Tourism-Related Complaint Count:** Monthly count of 311 complaints classified as tourism-related.

To normalize the complaint data for cities of different sizes and reporting volumes, a **tourism complaint share** was computed as:

$$\text{Tourism Complaint Share} = \frac{\text{Tourism-Related Complaints}}{\text{Total 311 Complaints}}$$

This metric represents the proportion of resident complaints likely influenced by tourism activity rather than overall service usage. All variables were checked for outliers and unrealistic values prior to analysis.

4. Quantitative Analysis

The quantitative analysis uses descriptive correlation analysis, cross city comparative analysis, and seasonal time-series analysis to examine how tourism activity relates to resident service burden. These methods were selected to identify both overall associations and differences in how cities experience tourism pressure over time.

Monthly tourism intensity values were compared to tourism complaint share values for each city. Scatter plots were used to visually assess whether increases in tourism demand corresponded to gradual or accelerated increases in resident complaints. To support visual findings, city-level correlation measures were calculated to quantify the strength and direction of the association between tourism intensity and tourism complaint share. These

statistics were used for descriptive comparison across cities rather than causal inference.

Time-series analysis was also conducted to examine how tourism activity and resident complaint shares evolved over time within each city, highlighting seasonal trends and periods of elevated tourism pressure.

5. Visualization and Tools

All data processing and analysis was conducted using Python. The pandas and numpy libraries were used for data cleaning, aggregation, and numerical calculations, the matplotlib library was used to generate scatter plots and time-series visualizations, and visual outputs were designed to clearly communicate trends and differences across cities in an accessible and interpretable format.

6. Ethical Considerations and Limitations

All data used in this study are publicly available and anonymized, and do not contain any personally-identifiable information. The analysis focuses on aggregate trends rather than individual behavior, and it is exploratory and descriptive in nature instead of attempting causal inference.

There are limitations to this method, such as use of search interest as a proxy for tourism rather than direct visitor counts and potential variation in how cities categorize 311 complaints. However, consistent preprocessing and within-city comparisons help to reduce these effects.



Results

4.1 Analysis of the Data Collected

This section discusses the findings from the combined analysis of tourism activity and resident service burden across cities. Using Google Trends, tourism activity was measured using the monthly search interest, and the resident service burden was measured using the 311 service request data.

Overall, the results showed that tourism activity and tourism-related complaint share had a positive association, which implies that periods of higher tourism demand coincided with high proportions of resident complaints of to tourism spillovers. However, the strength and consistency of this relationship varied by city, which suggests that cities experience and manage tourism pressure differently.

In some cities, increased tourism intensity was associated with a gradual increase in complaints, which showed that resident burden rose proportionally alongside tourism demand. In other cities, the complaint shares increased more drastically during high-tourism periods, which suggests the presence of saturation effects, where the additional tourism demand results in a disproportionately higher resident service strain.

Further analysis using time-series revealed seasonal patterns in tourism activity and complaint shares. The tourism intensity typically peaked during warmer months and peak travel seasons, aligning with increases in resident complaints. Although the overall trends were similar across cities, the size of these seasonal variations were different, emphasizing the need for city-specific context.

Correlation analysis also supported these observations by showing that the relationship between tourism activity and tourism-related complaint share was not the same across all cities. Some cities had stronger correlations, while others showed weaker or more varying relationships, suggesting differences in infrastructure capacity, tourism management strategies, and resident tolerance.

4.2 Monthly Aggregated Complaint Metrics

To understand overall patterns in the data, monthly totals of resident-filed 311 complaints and tourism-related complaints were calculated for each city. From

these totals, the share of complaints related to tourism was computed for each month.

The table below presents sample rows from the monthly aggregated dataset for Boston, showing total 311 calls, tourism-related calls, tourism complaint share, and tourism intensity values. As shown, tourism-related complaints make up a noticeable portion of all service requests, and this share varies across months, providing a clear starting point for examining how changes in tourism activity relate to resident complaints over time.

City	Month	Total 311 Calls	Tourism-Related Calls	Tourism Complaint Share	Tourism Intensity
Boston	2019-01-01	19,298	5,627	0.292	54.33
Boston	2019-02-01	16,846	4,887	0.290	59.00
Boston	2019-03-01	24,507	6,571	0.268	65.33
Boston	2019-04-01	20,378	6,389	0.314	69.33
Boston	2019-05-01	22,540	7,044	0.313	68.67

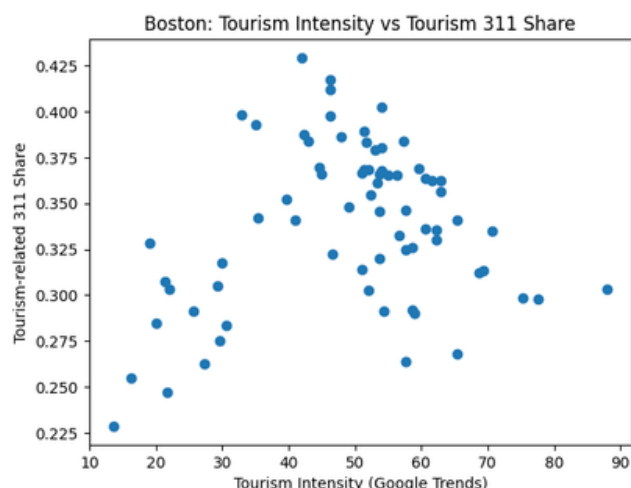
4.3 Tourism Intensity and Complaint Share Relationships

Scatter plots help examine the relationship between tourism intensity and tourism-related complaint share for each city.

As shown in the scatter plots, these three cities display an upward trend in which higher tourism intensity corresponds to higher tourism-related complaint shares. However, the slope and dispersion of points differ across cities, indicating heterogeneous responses to tourism demand. In some cities, complaint shares increase gradually with tourism activity, while in others, higher-intensity periods are associated with more pronounced increases in resident service burden.

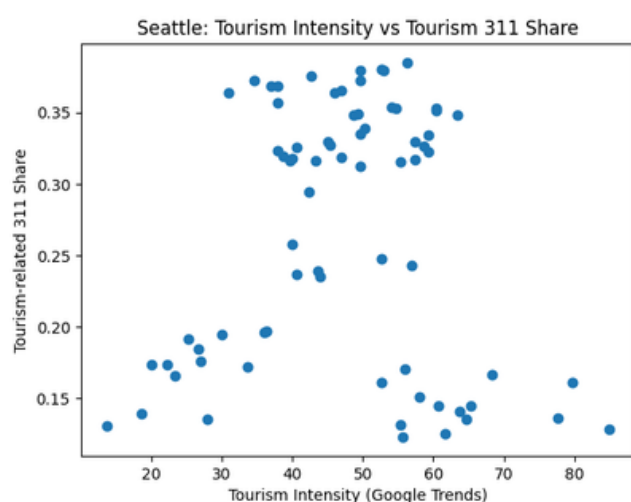
Boston

In Boston, the relationship between tourism intensity and tourism-related complaint share is moderate and positive, as complaints increase gradually as tourism activity increases. This pattern suggests that while tourism contributes to resident service burden, the city's infrastructure and service capacity may absorb increased demand without sharp saturation effects.



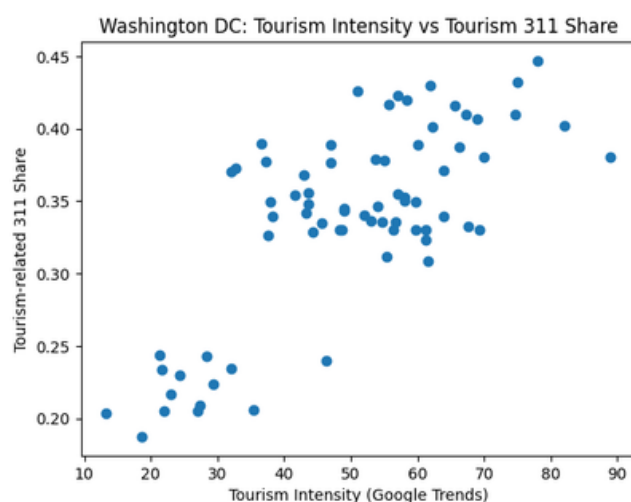
Seattle

Seattle exhibits a more clustered relationship between tourism intensity and complaint share, with some noticeable seasonal variation. While higher tourism levels are associated with increased complaints, the dispersion of points suggests that other factors, such as event-driven tourism or local enforcement practices, may influence resident reporting behavior.



Washington DC

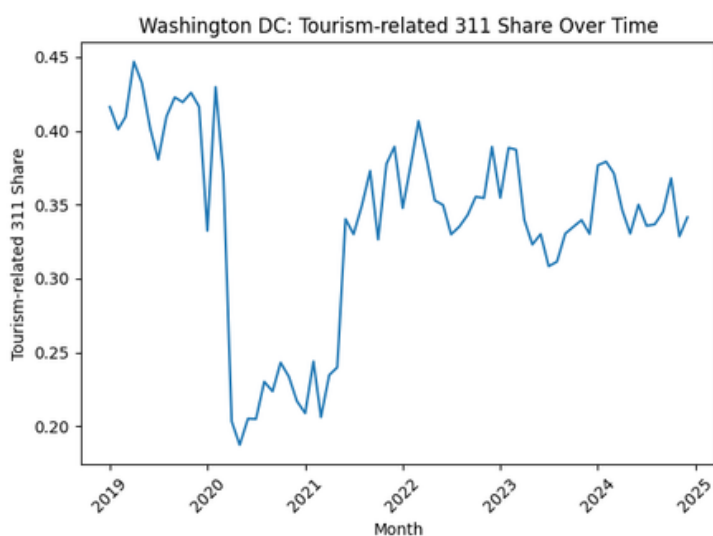
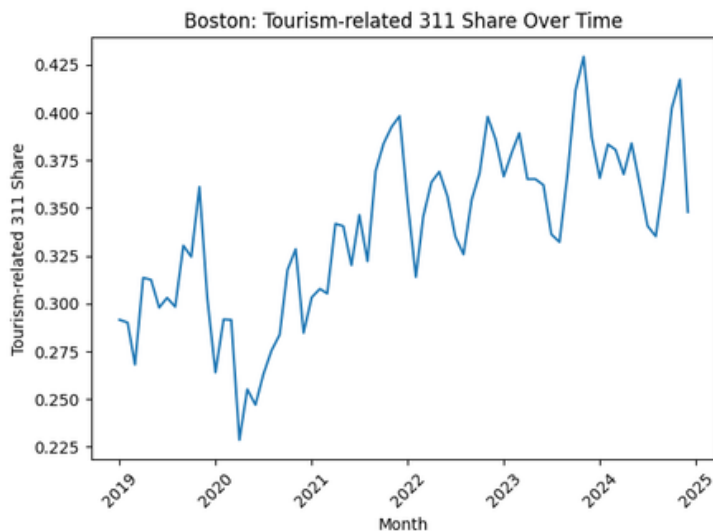
Washington, DC displays a steeper relationship at higher tourism intensity levels, where increases in tourism demand correspond to sharper increases in tourism-related complaint share. This pattern is consistent with a potential tourism saturation threshold, where additional tourism places disproportionate strain on resident services.



4.4 Temporal Patterns and Seasonality

Over time, the data shows clear seasonal patterns in both tourism activity and resident complaints. As shown in the diagrams below, tourism intensity increases during warmer months and popular travel seasons.

These higher tourism periods often line up with increases in the share of tourism-related complaints, suggesting that tourism impacts on residents are stronger during peak travel times. Even though this pattern appears in all cities, the size and consistency of the increases differ by city, which shows that cities experience tourism pressure differently. The graphs also show a noticeable drop in tourism activity between 2019 and 2021, which is likely related to the COVID-19 pandemic.



4.5 Correlation Analysis Across Cities

City	level_1	n_311_calls	n_tourism_calls	tourism_call_share	tourism_intensity
Boston	n_311_calls	1	0.78	0.03	0.4
Boston	n_tourism_calls	0.78	1	0.64	0.43
Boston	tourism_call_share	0.03	0.64	1	0.22
Boston	tourism_intensity	0.4	0.43	0.22	1
Seattle	n_311_calls	1	0.92	0.7	0.45
Seattle	n_tourism_calls	0.92	1	0.91	0.21
Seattle	tourism_call_share	0.7	0.91	1	0.01
Seattle	tourism_intensity	0.45	0.21	0.01	1
Washington DC	n_311_calls	1	0.81	0.29	0.53
Washington DC	n_tourism_calls	0.81	1	0.79	0.79
Washington DC	tourism_call_share	0.29	0.79	1	0.74
Washington DC	tourism_intensity	0.53	0.79	0.74	1

To further examine the relationship between tourism activity and resident service burden, Pearson correlation coefficients (r values) were calculated between key tourism indicators and 311 call metrics for each city. The table above summarizes correlations among total 311 calls, tourism-related 311 calls, tourism call share, and the constructed tourism intensity index for Boston, Seattle, and Washington, DC. (Code in Appendix D)

Across all three cities, total 311 calls and tourism-related 311 calls are strongly positively correlated, with coefficients of 0.78 in Boston, 0.92 in Seattle, and 0.81 in Washington, DC, suggesting that periods of higher overall service demand also tend to coincide with increased tourism-related complaints.

However, the relationship between tourism intensity and resident service burden varies notably by city. In Boston, tourism intensity shows a moderate positive correlation with total 311 calls (0.40) and tourism-related calls (0.43), but a weaker relationship with tourism call share (0.22), indicating that tourism contributes to service demand without dominating complaint composition.

In Seattle, tourism intensity exhibits little to no correlation with tourism call share (0.01) and only a weak relationship with tourism-related calls (0.21), despite strong correlations among tourism-related variables themselves. This suggests that tourism-related complaints in Seattle scale with overall activity but do not significantly increase their share of total 311 calls.

In contrast, Washington, DC displays consistently strong correlations between tourism intensity and both tourism-related calls (0.79) and tourism call share (0.74), indicating a tighter coupling between tourism activity and resident service burden. This pattern aligns with DC's role as a national tourism destination, where fluctuations in visitor volume more directly affect resident-facing city services.

These results highlight that while tourism activity is associated with increased service demand across all cities, the strength and nature of this relationship are highly context-dependent, reinforcing the need for city-specific tourism management and policy responses.



Conclusion

This TSA Data Science and Analytics project explored the relationship between tourism activity and resident service burden, based on how they are associated and whether this relationship differs across cities. By combining Google Trends tourism data with resident-filed 311 service requests, the analysis offers insight into how changes in tourism demand relate to urban quality-of-life impacts.

The results show that higher tourism activity is generally linked to increases in tourism-related resident complaints. However, this relationship varies by city, indicating that tourism pressure does not affect all urban areas in the same way. Some cities appear better able to accommodate tourism demand with relatively smaller increases in resident complaints, while others experience sharper rises during periods of high tourism activity. Seasonal patterns and city-specific differences suggest that local conditions such as infrastructure capacity, service availability, and tourism management practices play an important role in shaping resident experiences. Cities with stronger planning and public service capacity may be more resilient to increased tourism, while cities facing rapid tourism growth may experience greater strain.

Overall, these findings highlight that tourism growth can contribute to resident service burden, but the scale of its impact depends heavily on local context. This underscores the importance of using city-specific data to inform tourism management strategies rather than relying on uniform approaches across different urban environments.



Next Steps

There are several ways to build on this analysis in future work:

1. Expanding the study to include additional cities would improve comparisons across different urban settings. Including smaller tourist destinations or cities outside the United States could provide further insight into how city size and tourism scale influence resident service burden.
2. Tourism activity could be measured using additional indicators such as hotel occupancy rates, short term rental activity, airport passenger counts, or other publicly available data. Using multiple measures would strengthen the analysis and reduce reliance on a single proxy.
3. The method used to identify tourism related complaints could be improved by applying more advanced text analysis techniques. This would allow for more accurate classification of complaints and reduce potential misclassification.
4. Future work could explore how specific local policies such as noise regulations, crowd management efforts, or limits on short term rentals affect tourism related complaints over time. This type of analysis could help cities better balance tourism growth with resident quality of life.



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Appendix A : Google Search Trends Data Pipeline (Code)

Appendix A documents the Python code used to collect and process Google Search Trends data to construct monthly tourism intensity measures.

Github Link:

[https://github.com/Snet101/TSA_Tourism/blob/main/TourismSearchTrendsAppendixA\(NCTSA\).ipynb](https://github.com/Snet101/TSA_Tourism/blob/main/TourismSearchTrendsAppendixA(NCTSA).ipynb)

Appendix B: 311 Complaint Data Processing Pipeline (Code)

Appendix B presents the data cleaning and aggregation pipeline used to identify tourism-related 311 complaints and compute monthly complaint metrics.

Github Link:

[https://github.com/Snet101/TSA_Tourism/blob/main/311ProcessingAppendixB\(NCTSA\).ipynb](https://github.com/Snet101/TSA_Tourism/blob/main/311ProcessingAppendixB(NCTSA).ipynb)

Appendix C: Monthly Aggregated Complaint Metrics (Sample Year)

Appendix C presents representative monthly aggregated complaint metrics for each city using data from 2019. For readability, one year of data is shown per city. All analyses in this project were conducted using the full dataset.



Table C1. Boston Monthly Aggregated Complaint Metrics (2019)

Month	Total 311 Calls	Tourism-Related Calls	Tourism Complaint Share	Tourism Intensity
2019-01	19,298	5,627	0.292	54.33
2019-02	16,846	4,887	0.290	59.00
2019-03	24,507	6,571	0.268	65.33
2019-04	20,378	6,389	0.314	69.33
2019-05	22,540	7,044	0.313	68.67
2019-06	23,545	7,015	0.298	77.67
2019-07	24,278	7,358	0.303	88.00
2019-08	26,486	7,903	0.298	75.33
2019-09	24,796	8,192	0.330	62.33
2019-10	21,324	6,920	0.325	57.67
2019-11	16,942	6,118	0.361	53.33
2019-12	18,082	5,477	0.303	52.00

Table C2. Seattle Monthly Aggregated Complaint Metrics (2019)

Month	Total 311 Calls	Tourism-Related Calls	Tourism Complaint Share	Tourism Intensity
2019-01	14,957	1,879	0.126	61.67
2019-02	11,502	1,417	0.123	55.67
2019-03	15,510	2,108	0.136	64.67
2019-04	15,913	2,240	0.141	63.67
2019-05	16,440	2,389	0.145	65.33
2019-06	15,977	2,579	0.161	79.67
2019-07	19,805	2,553	0.129	85.00
2019-08	17,433	2,380	0.137	77.67
2019-09	16,272	2,712	0.167	68.33
2019-10	14,811	2,526	0.171	56.00
2019-11	12,298	1,987	0.162	52.67
2019-12	12,950	1,954	0.151	58.00

Table C3. Washington DC Monthly Aggregated Complaint Metrics (2019)

Month	Total 311 Calls	Tourism-Related Calls	Tourism Complaint Share	Tourism Intensity
2019-01	28,914	12,035	0.416	65.67
2019-02	25,790	10,343	0.401	62.33
2019-03	30,702	12,572	0.409	74.67
2019-04	31,633	14,135	0.447	78.00
2019-05	34,934	15,107	0.432	75.00
2019-06	33,711	13,548	0.402	82.00
2019-07	36,285	13,807	0.381	89.00
2019-08	35,480	14,537	0.410	67.33
2019-09	30,372	12,839	0.423	57.00
2019-10	29,940	12,557	0.419	58.33
2019-11	23,154	9,860	0.426	51.00
2019-12	26,140	10,884	0.416	55.67

Appendix D: Correlation Analysis Code

Appendix D includes the Python code used to examine how tourism activity and 311 service calls change together over time using monthly aggregated data for Boston, Seattle, and Washington, DC.

Github Link:

[https://github.com/Snet101/TSA_Tourism/blob/main/CorrelationTableAppendixD\(NCTSA\).ipynb](https://github.com/Snet101/TSA_Tourism/blob/main/CorrelationTableAppendixD(NCTSA).ipynb)